Topic 6: Optionals and Optional Handling

Task 1: User Profile with Missing Data

```
struct Address {
   var city: String?
}

struct User {
   var name: String
   var age: Int?
   var address: Address?
}

func describe(_ user: User) -> String {
   guard let age = user.age else {
      return "\(user.name): Age is unknown"
   }

   let city = user.address?.city ?? "unknown city"
   return "\(user.name), Age: \(age), City: \(city)"
}

let user1 = User(name: "Gayathiri", age: 22, address: Address(city: "Chennai"))
let user2 = User(name: "Dhanuja", age: nil, address: Address(city: "Coimbatore"))
let user3 = User(name: "akchu", age: 20, address: Address(city: nil))

print(describe(user1))
print(describe(user2))
print(describe(user3))

let result1 = Int("42")
let result2 = Int("abc")

print(result1 as Any)
print(result2 as Any)
```

output:

Output

```
gayathiri, Age: 22, City: Chennai
dhanuja: Age is unknown
akchu, Age: 20, City: unknown city
Optional(42)
nil
```

Task 2: Safe Number Parsing

```
let values = ["10", "x", "25", "", "7.5", "40"]

let numbers = values.compactMap { Int($0) }
let sum = numbers.reduce(0, +)
let invalidCount = values.count - numbers.count
let doubled = numbers.first.map { $0 * 2 }

print("Valid integers:", numbers)
print("Sum:", sum)
print("Invalid entries:", invalidCount)
print("First valid number doubled:", doubled as Any)
```

output:

Output

```
Valid integers: [10, 25, 40]
Sum: 75
Invalid entries: 3
First valid number doubled: Optional(20)
```

Task 3: Remove the Crashes

```
let scores = ["Gayathiri": 90, "vaishu": 72]

if let anu = scores["anu"] {
    if let number = Int("12a") {
        print(anu + number)
    } else {
        print("Invalid number")
    }
} else {
    print("anu's score not found")
}

if let number = Int("12a") {
    print("Number:", number)
} else {
    print("Cannot convert 12a to an integer")
}
```

Output:

Output

```
anu's score not found
Cannot convert 12a to an integer
```